

Geography Ch-3 (Advanced) — Seismic Hazard Zones of India

Advanced / India-applied | GS-I & GS-III (Disaster) | Source: D.R. Khullar (India: A Comprehensive Geography)

Complements Must-Do Ch-3 (Vulcanism & Earthquakes) with India's seismic risk map

Extra CA: BIS seismic zonation & Himalayan plate-collision seismicity

Facts from official sources | Inferences clearly marked | No fabricated data

HIGH

ADVANCED — India's Earthquake Zones, Plate Collision & Liquefaction (D.R. Khullar)

■ NEWS TRIGGER

India's seismic risk is set by the ongoing northward push of the Indian Plate (~5 cm/yr) into Eurasia; the National Center for Seismology maps quakes against the BIS seismic-zone map (Zone II–V), with the Himalaya, NE India and Kutch in the highest-risk Zone V.

■ INTRO & ORIGIN

Khullar divides India into seismological zones by earthquake frequency and intensity. The **Himalayan fold-mountain belt** is the most earthquake-prone; the **Peninsular shield** is relatively stable (with exceptions like Latur and Koyna).

*The root cause is **plate tectonics**: the Indian Plate's northerly drift and collision with the Asian Plate built the Himalaya and keeps accumulating strain — released as earthquakes along thrust faults.*

■ KEY DATA

BIS Zone	Risk (MSK intensity)	Representative areas
Zone V	Very high (IX+)	NE India, Kashmir/Himachal/Uttarakhand Himalaya, Kutch, Andaman
Zone IV	High (VIII)	Delhi-NCR, remaining Himalayan foothills, Indo-Gangetic belt
Zone III	Moderate (VII)	Parts of Peninsula, Gujarat, west coast
Zone II	Low (VI)	Stable interior Peninsula

■ STATIC THEORY

- India's most earthquake-prone belt is the **Himalayan zone** (Kashmir & Western Himalayas, Central Himalayas, NE India) — the plate-collision front.
- The **Indo-Gangetic basin** is high-risk because thick alluvium amplifies shaking; **Kutch (Rann)** is an intraplate high-risk pocket (2001 Bhuj quake).
- The **Peninsular shield** is presumably a stable landmass with few quakes — but Latur (1993) and Koyna (reservoir-induced) are notable exceptions.
- Liquefaction**: during strong quakes, loose water-saturated uniform sands lose strength and behave like a liquid — buildings sink or tilt (severe in Bhuj 2001).
- Intensity in India is mapped on the **MSK scale** (damage felt), while magnitude uses the Richter scale.

■ MUST-KNOW FACTS

- BIS seismic zones: II (low) to V (very high) — IS:1893.
- Zone V (highest): NE India, Himalaya (J&K/HP/UK), Kutch, Andaman & Nicobar.
- Delhi-NCR = Zone IV; interior Peninsula = Zone II.

■ UPSC TRAPS

- ✗ The Peninsular shield is completely aseismic.
- ✓ It is RELATIVELY stable but not immune — Latur and Koyna are significant intraplate quakes.
- ✗ Highest Indian seismic zone is Zone I.

4. Cause: Indian Plate ~5 cm/yr collision with Eurasia → Himalayan thrust quakes. 5. Liquefaction = saturated loose sand losing strength (Bhuj 2001). 6. Peninsular exceptions: Latur (1993), Koyna (reservoir-induced).	✓ The BIS map runs Zone II–V; Zone V is the HIGHEST-risk (there is no Zone I in the current map). ✗ Alluvial plains are safest during quakes. ✓ Thick alluvium AMPLIFIES shaking (Indo-Gangetic plains are high-risk), and can liquefy.
■ MAINS ANGLE <i>Explain why the Indian-Eurasian collision makes the Himalaya a Zone-V seismic belt, and how BIS zonation must drive building codes, retrofitting and land-use in high-risk Indian cities.</i>	■ STUDY LINK Geography → India → Natural Hazards → Seismic Zones (Khullar, Natural Hazards & Disasters)

End of Register Notes — UPSC Agent / Copilot CLI